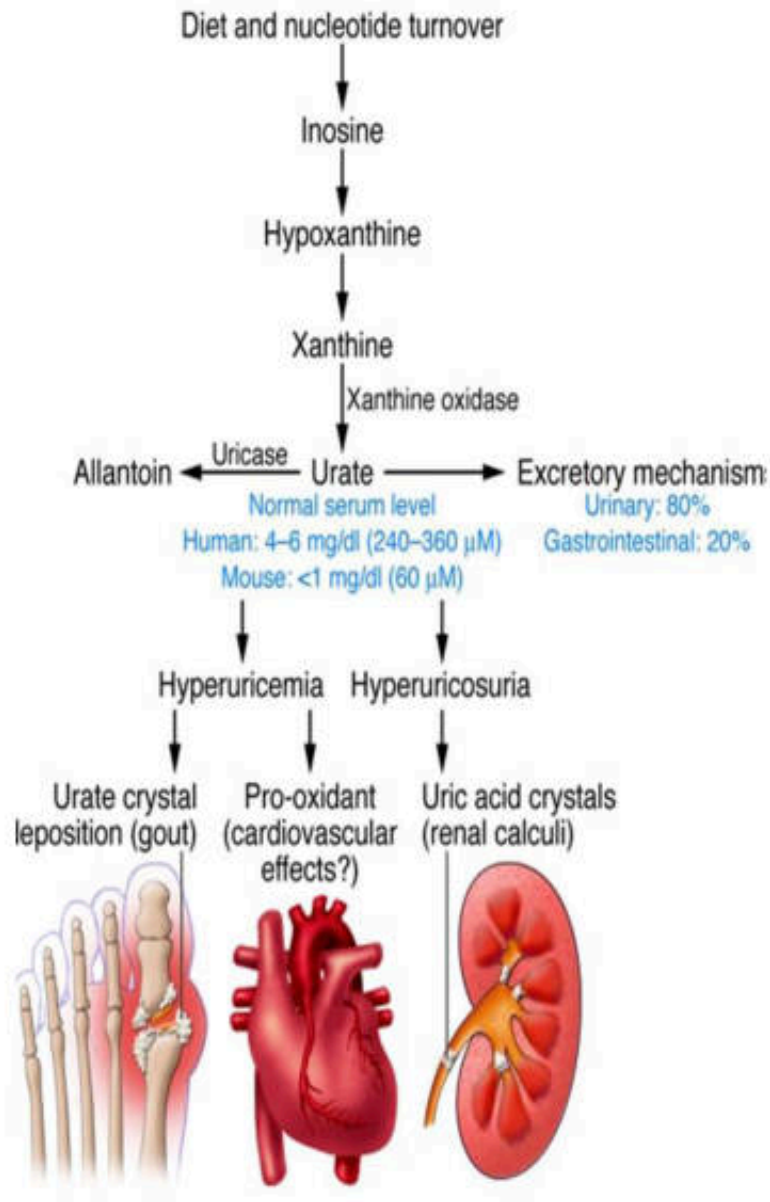


Pharmacology of drugs used for treatment of acute and chronic gout

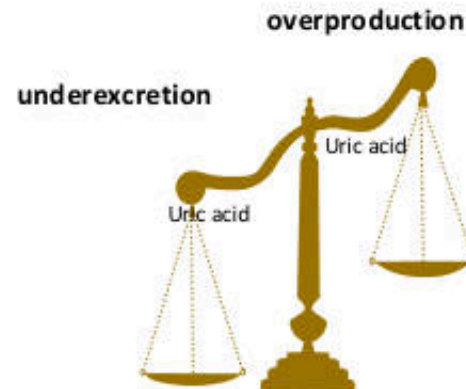
Assoc. Prof. Dr. Vilasinee Hirunpanich Sato
Department of Pharmacology
Faculty of Pharmacy
Mahidol University



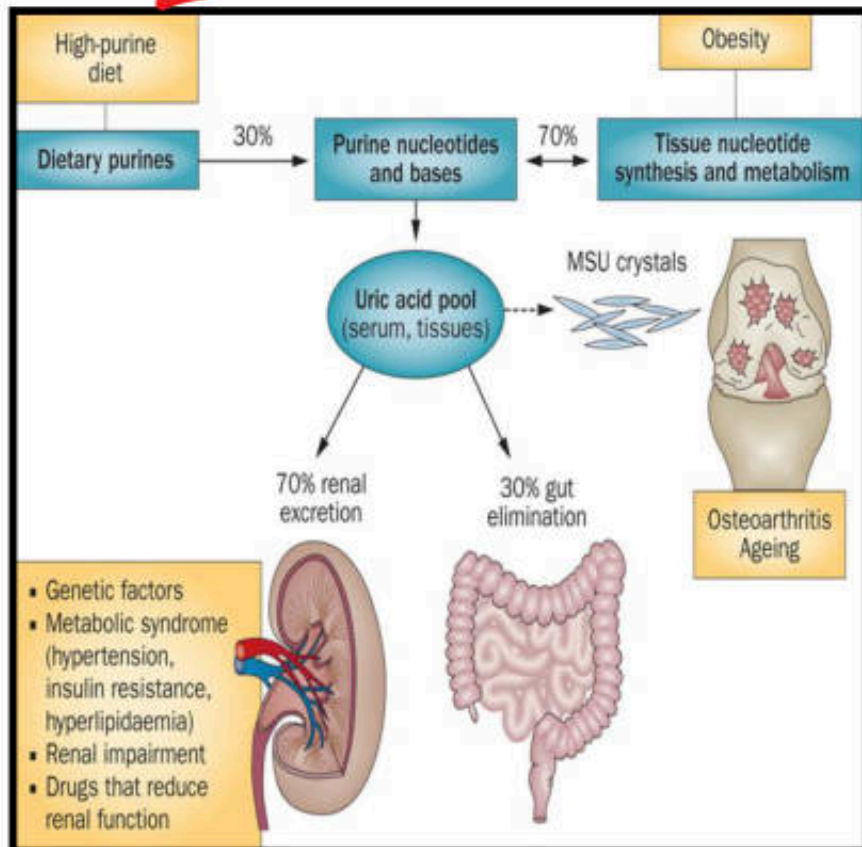
Hyperuricemia



- Serum urate concentration > 6.8 mg/dl (408 μmol/L)
- Guideline for the management of gout in Thailand (2012):
 - Male > 7 mg/dl, female > 6 mg/dl
- Epidemiology: Aging (male > 65 yr.: 7% and women > 86 yr.: 3%)
- Hyperuricemia is strongly associated with the degree of gout, cardiovascular, metabolic syndrome, kidney diseases, DM type 2



สาเหตุของระดับกรดยูริกในเลือดสูง



RISK FACTORS FOR HYPERURICEMIA

Demographics

- Older age (over 40 years in men and over 60 years in women)
- Male sex

Diet and Alcohol

- High intake of red meat, organ meats, shellfish, and dried beans
- Heavy alcohol use (especially beer)

Medications

- | | |
|----------------|---|
| • Aspirin | • Tacrolimus |
| • Cyclosporin | • Warfarin |
| • Ethambutol | • Diuretics ("water" pills) |
| • Levodopa | • Drugs used to treat cancer |
| • Niacin | • Overuse of laxatives used to treat constipation |
| • Pyrazinamide | |

Treatment aims of gout

1. To relieve acute gouty attacks

1. **NSAIDs**: all can be used except aspirin (low dose (<2.6 g/d) reduce urate excretion)

2. **Corticosteroids**: oral, i.m., intra-articular (i.a.)

3. **Colchicine**

4. **IL-1 inhibitors**: Canakinumab

2. To reduce urate crystal in chronic gout

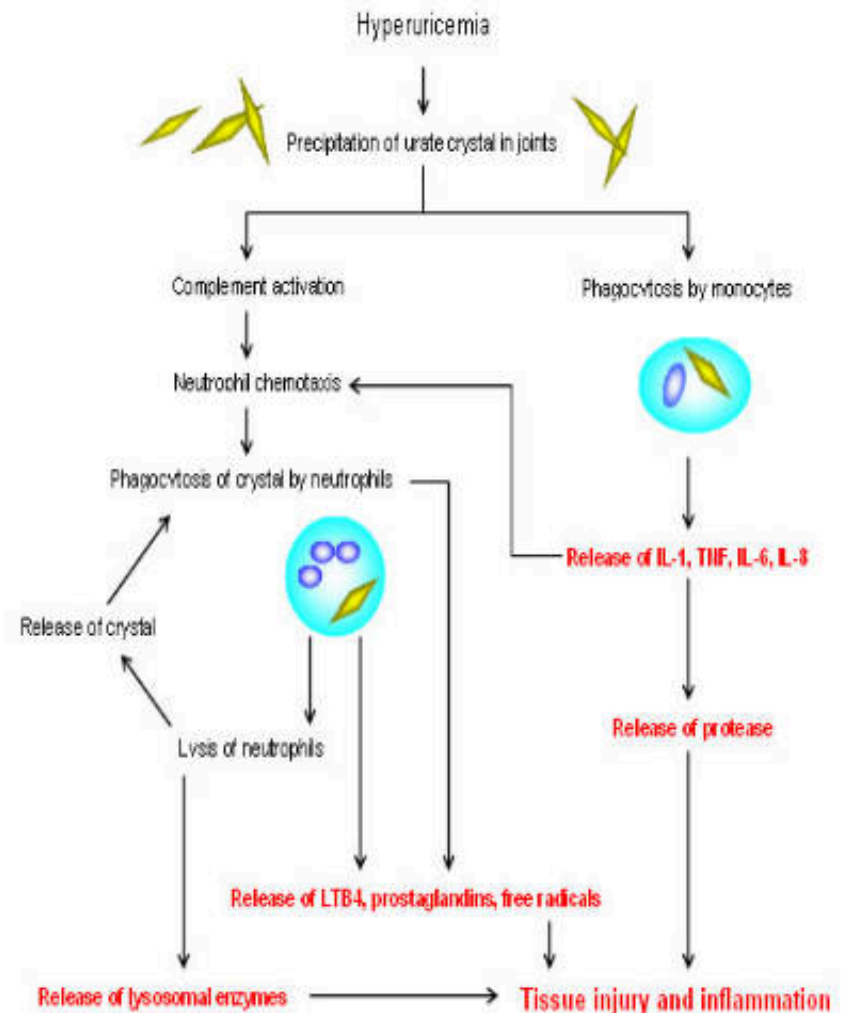
1. **Inhibit uric acid synthesis** (Urostatic)

: Allopurinol, Febuxostat

2. **Increase uric acid excretion** (Uricosuric):
Probenecid, Sulphinpyrazole, Lesinurad

3. **Uricase**: Plegloticase

Acute gouty attacks



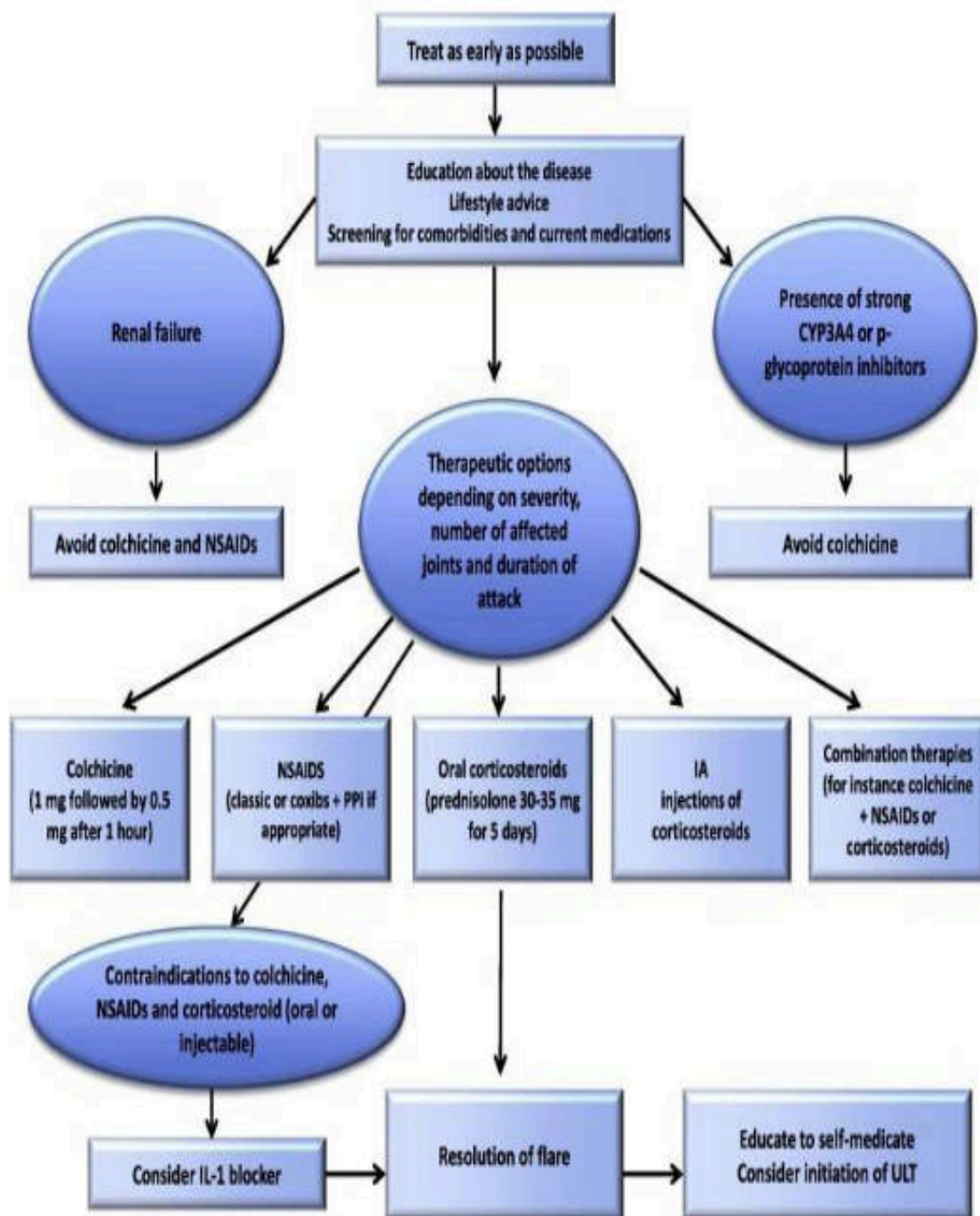
1. Drugs used for acute gouty attacks

1. Colchicine

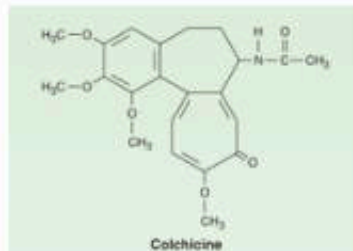
2. NSAIDs

3. Corticosteroids: oral, i.m., intra-articular (i.a.)

4. IL-1 inhibitors



1. Colchicine



Mechanism of actions:

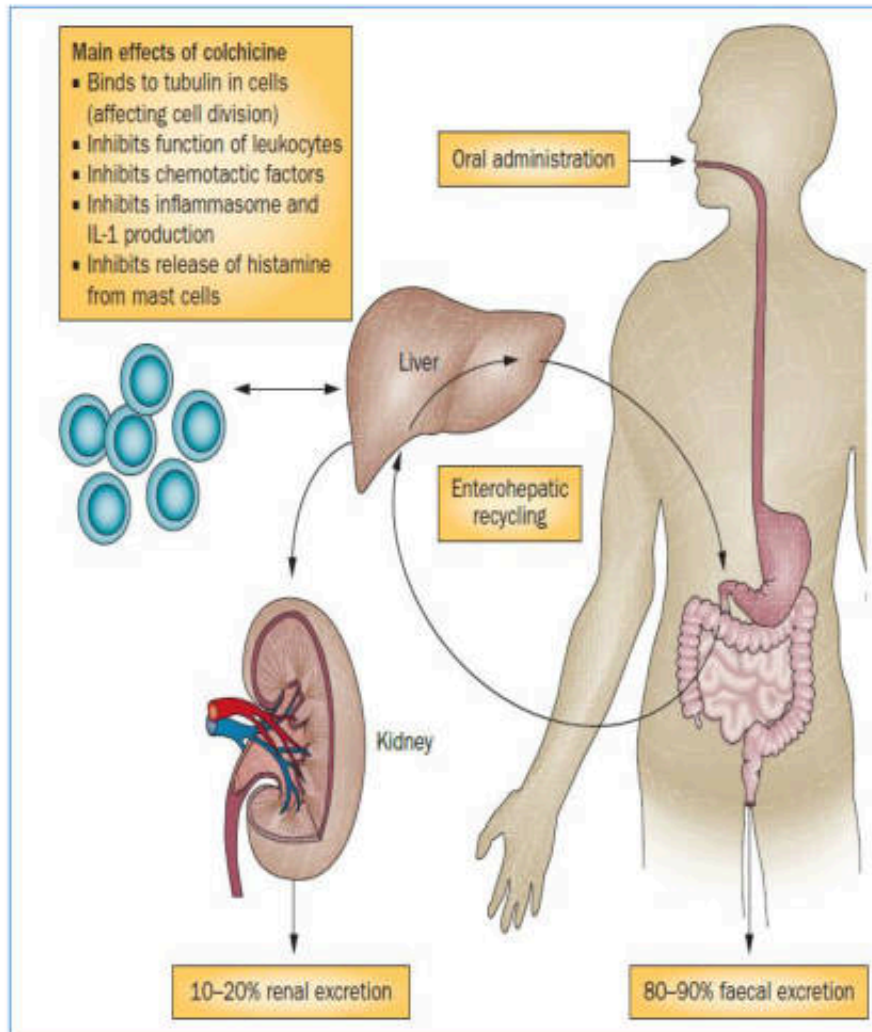
- binding to the intracellular protein tubulin → preventing polymerization into microtubules → inhibition of leukocyte migration and phagocytosis
- Inhibit the formation of leukotriene B₄ and IL-1 β

Dosage:

- **Gouty attack:** a regimen of 1.2 mg followed by a single 0.6-mg oral dose
- **Prophylaxis:** 0.6 mg 1-3/day

Narrow therapeutic index

- Colchicine is a P-gp and CYP3A4 substrates → avoid to use with CYP3A4/P-gp inhibitors (drug interaction)
- Eliminate by renal excretion
- **Major ADRs:** GI side effects (diarrhea 23%, N/V (4-17%), stomachache), fatigue and headache, neuritis, myopathy, alopecia, bone marrow suppression
- Adjust dose in patient with renal disease



Drug Interactions

Drug	Noted or Anticipated Outcome	Clinical Comment	
Strong CYP3A4 Inhibitors			
atazanavir, clarithromycin, indinavir, itraconazole, ketoconazole, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin	Significant increase in colchicine plasma levels ¹ ; fatal colchicine toxicity has been reported with clarithromycin, a strong CYP3A4 inhibitor. Similarly, significant increase in colchicine plasma levels is anticipated with other strong CYP3A4 inhibitors.	Gout Flares 0.6 mg (1 tablet) x 1 dose, followed by 0.3 mg (half tablet) 1 hour later. Dose to be repeated no earlier than 3 days.	FMF Maximum daily dose of 0.6 mg (may be given as 0.3 mg twice a day)
Moderate CYP3A4 Inhibitors			
amprenavir, aprepitant, diltiazem, erythromycin, fluconazole, fosamprenavir, grapefruit juice, verapamil	Significant increase in colchicine plasma concentration is anticipated. Neuromuscular toxicity has been reported with diltiazem and verapamil interactions.	Gout Flares 1.2 mg (2 tablets) x 1 dose. Dose to be repeated no earlier than 3 days.	FMF Maximum daily dose of 1.2 mg (may be given as 0.6 mg twice a day)
P-gp Inhibitors			
cyclosporine, ranolazine	Significant increase in colchicine plasma levels ¹ ; fatal colchicine toxicity has been reported with cyclosporine, a P-gp inhibitor. Similarly, significant increase in colchicine plasma levels is anticipated with other P-gp inhibitors.	Gout Flares 0.6 mg (1 tablet) x 1 dose. Dose to be repeated no earlier than 3 days.	FMF Maximum daily dose of 0.6 mg (may be given as 0.3 mg twice a day)

FMP: familial Mediterranean fever

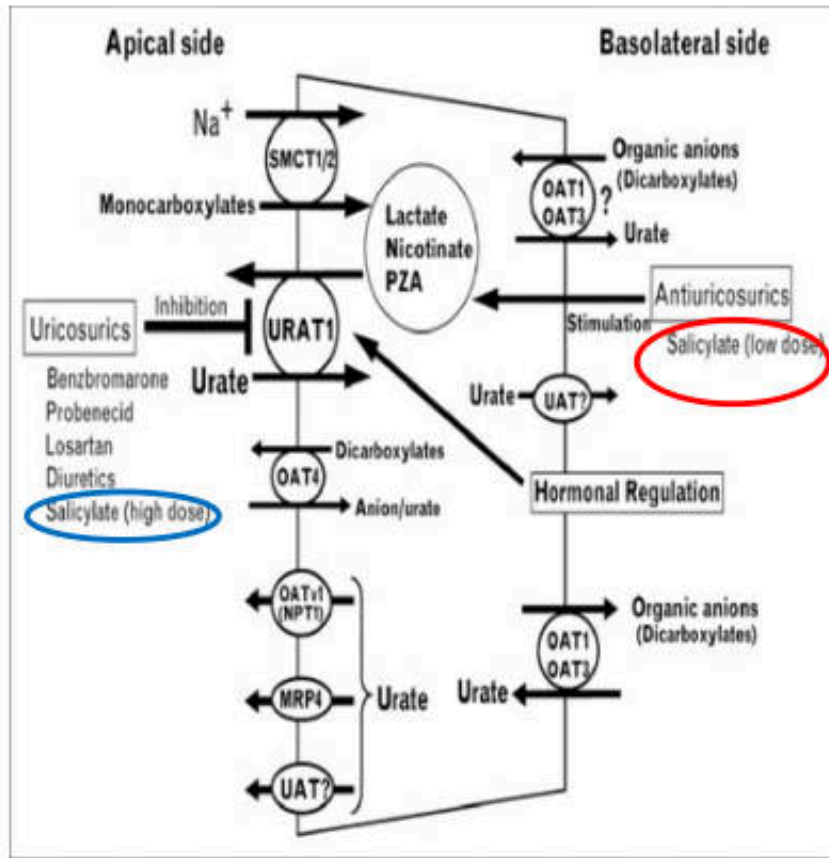
Indication of Colchicine

- FDA approval for gout prophylaxis and treatment of acute gouty flares**
 - Treatment of acute gout flare: 1.2 mg at the first sign of a gout flare followed by 0.6 mg one hour later.
 - Familial Mediterranean fever (FMP): 1.2 mg to 2.4 mg for adults and children over 12 years old; the daily dose gets administered in one or two doses.
- FDA (2023) approves colchicine for decreasing atherosclerotic cardiovascular events.**

บัญชียาหลักแห่งชาติ 2567

10.1.3. Drugs for treatment of gout and hyperuricaemia			
1	Colchicine	tab	ก
2	Allopurinol	tab	ก
3	Probenecid	film coated tab	ก
4	Benzbromarone	tab	ค
5	Febuxostat	tab (เฉพาะ 80 mg)	ง

2. Non-steroid anti-inflammatory drugs (NSAIDs)



- All NSAIDs can be used except aspirin...
 - Low dose aspirin (< 1-2 g/d): bind URAT1 transporter → ↑ serum uric acid
 - High dose aspirin (>3 g/d): uricosuric effect → ↓ serum uric acid
- Reduce pain in patients with acute gout and prevent gout flares during urate lowering therapy (ULTs)
- **Major ADRs:** GI side effects (may plus a proton pump inhibitor), Long-term use of higher doses can cause chronic renal insufficiency
- NSAIDs should be avoided in patients with severe renal impairment

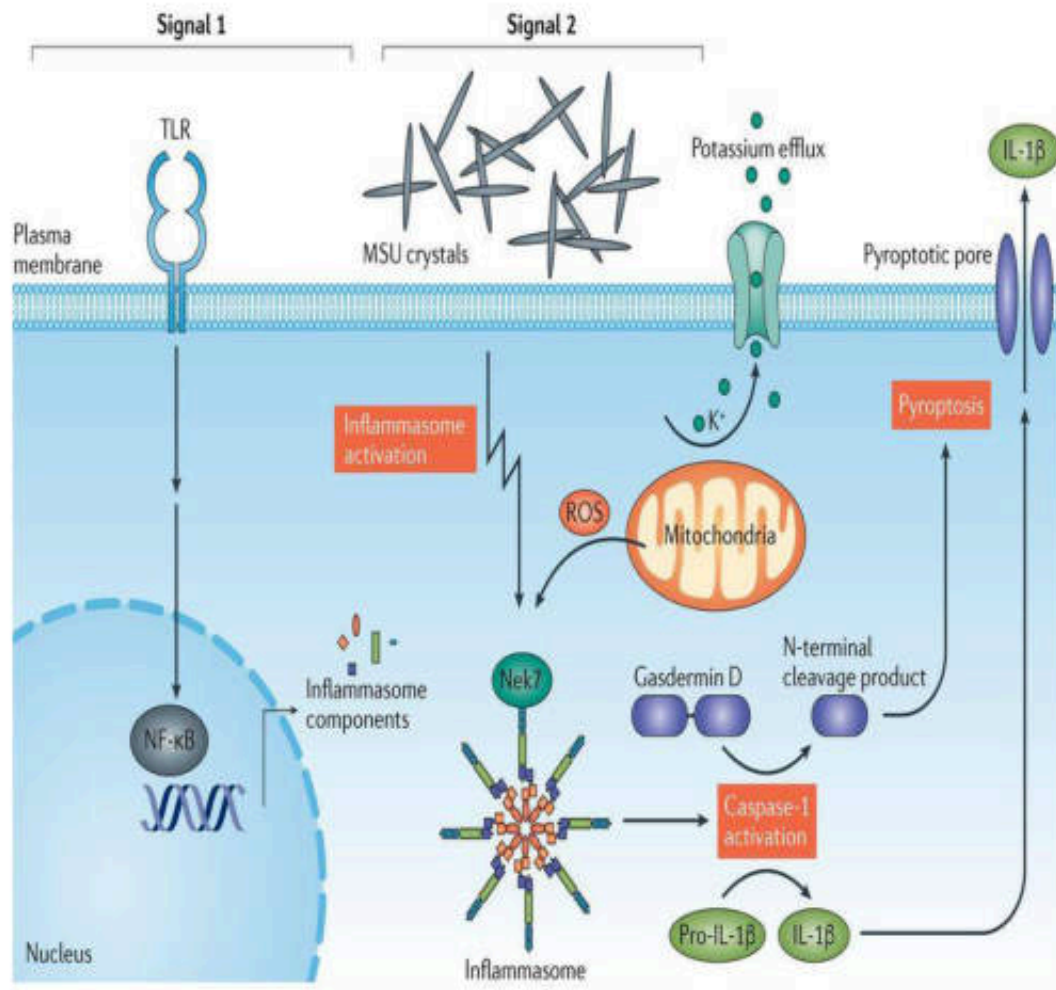
3. Corticosteroids

- European League Against Rheumatism (EULAR) guidelines recommendation of:
 - a short course (3–5 days) of an oral corticosteroid (30–35 mg/day prednisolone) in patients unable to tolerate NSAIDs or colchicine
- Intra-articular (i.a.) corticosteroid injections in acute gout can be helpful, particularly in for gout affecting a single joint
- Used as a combination of NSAIDs or colchicine
- **ADRs:** dysphoria, mood disorders, elevation of blood glucose levels, immune suppression, and fluid retention (long term used)

บัญชียาหลักแห่งชาติ 2567

6.3. Corticosteroids				
1	Dexamethasone		cap/tab (as base), sterile sol (as sodium phosphate or acetate)	ก
2	Hydrocortisone ชนิดเม็ดเป็นยากำพร้า		tab (as base), sterile pwdr (as sodium succinate), sterile susp (as acetate)	ก
3	Prednisolone		cap, tab	ก
4	Fludrocortisone acetate ยากำพร้า		tab	ข
5	Methylprednisolone		sterile pwdr/sterile susp (as hemisuccinate or sodium succinate or acetate)	ค
6	Triamcinolone acetonide		sterile susp	ค

4. Interleukin1 β -inhibitors



Nature Reviews | Rheumatology

- IL-1 β , are the key mediators of gouty inflammation
- The NLRP3 inflammasome is the major pathway by which monosodium urate (MSU) crystal-induced Inflammation $\rightarrow$ trigger the cellular inflammatory response
- IL-1 β inhibitors are considered to use in patients with frequent flares and contraindications to colchicine, NSAIDs and corticosteroids (oral and injectable)
- Example: Anakinra, Rilonacept, Canakinumab

แบบฝึกหัดทบทวนและทดสอบความเข้าใจ

- 1. ข้อใดสารกระตุ้นการอักเสบที่สำคัญที่สุดในการเกิด **acute gout**

- a. Prostaglandin (PG)
- B. TNF- β
- C. IL-1
- D. NO (nitric oxide)

- 2. ข้อใดเป็นกลไกการออกฤทธิ์ของ **colchicine**

- a. Inhibit prostaglandin production
- B. inhibit URAT1 transporter
- C. Bind tubulin and inhibit leucocytes
- D. Inhibit xanthine oxidase

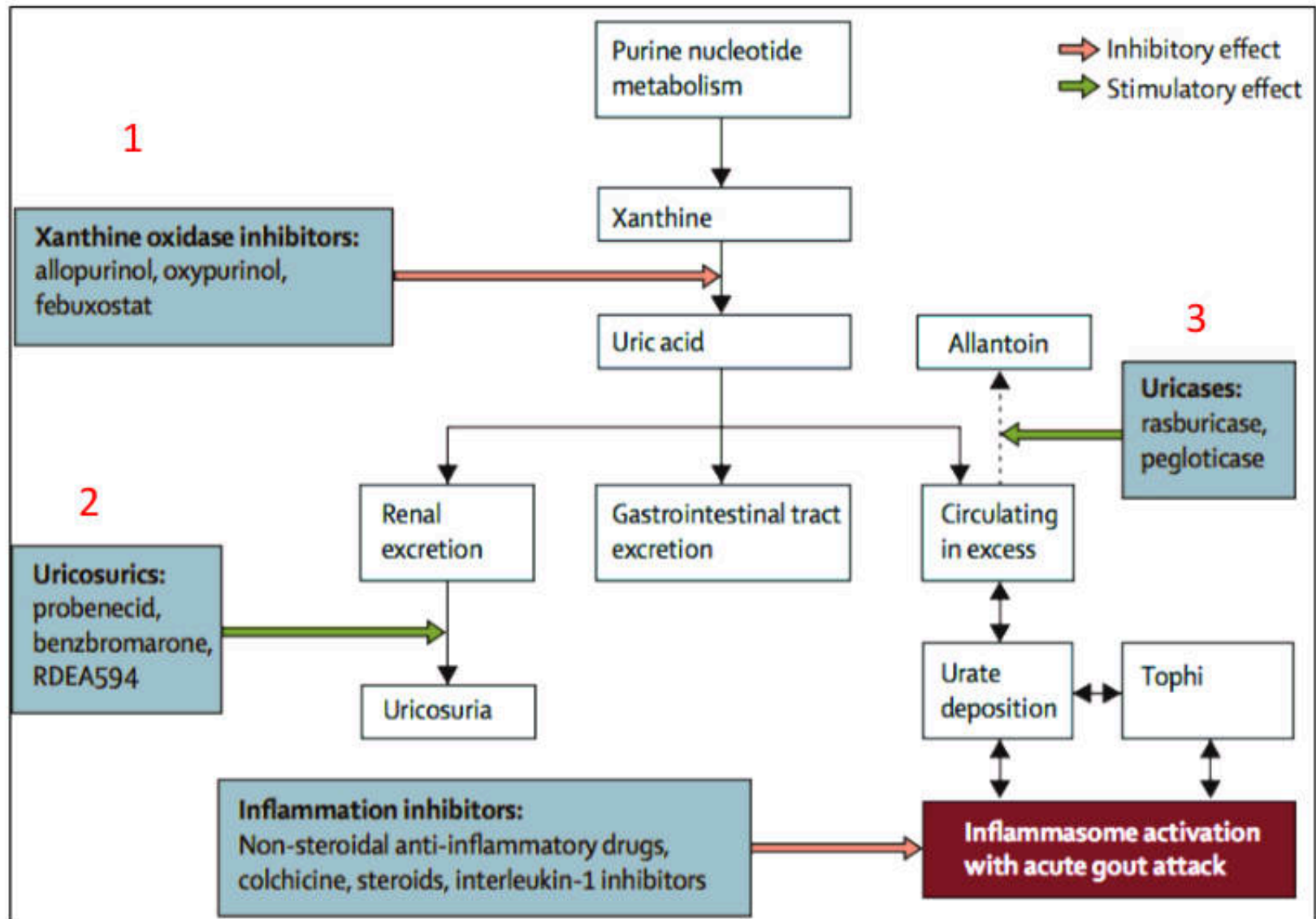
2. Chronic gout

Indications of urate lowering drugs

1. มีข้ออักเสบกำเริบ (gout flare) เป็นๆหายๆบ่อยกว่า 2-3 ครั้งต่อปี
2. มีข้ออักเสบเรื้อรัง; มีปุ่มโทฟัส (subcutaneous tophi)
3. มีความผิดปกติทางภาพรังสีกระดูกและข้อซึ่งเข้าได้กับโรคเกาต์
4. มีภาวะไตบกพร่อง และมีความจำเป็นต้องได้รับยาขับปัสสาวะอย่างต่อเนื่อง
5. มีหรือเคยมีอาการไตเสื่อมหรือไตอักเสบ uric acid nephropathy



Drugs used for treatment of Chronic gout



Current drugs for treatment of chronic hyperuricemia

1. Xanthine oxidase inhibitors (XOIs)

- First line for treatment of hyperuricemia
- Drugs: **Allopurinol**, **Febuxostat** (2009)

2. Uricosurics

- Second-line or alternative agents
- All recent guidelines support the combination with other agents (e.g. XOIs and uricosurics) when monotherapy is not effective
- Drugs: **Probenecid**, **Benzbromarone**, Sulfapyrazone, Lesinurad (withdrawn), Dotinurad (2023)

3. Uricase

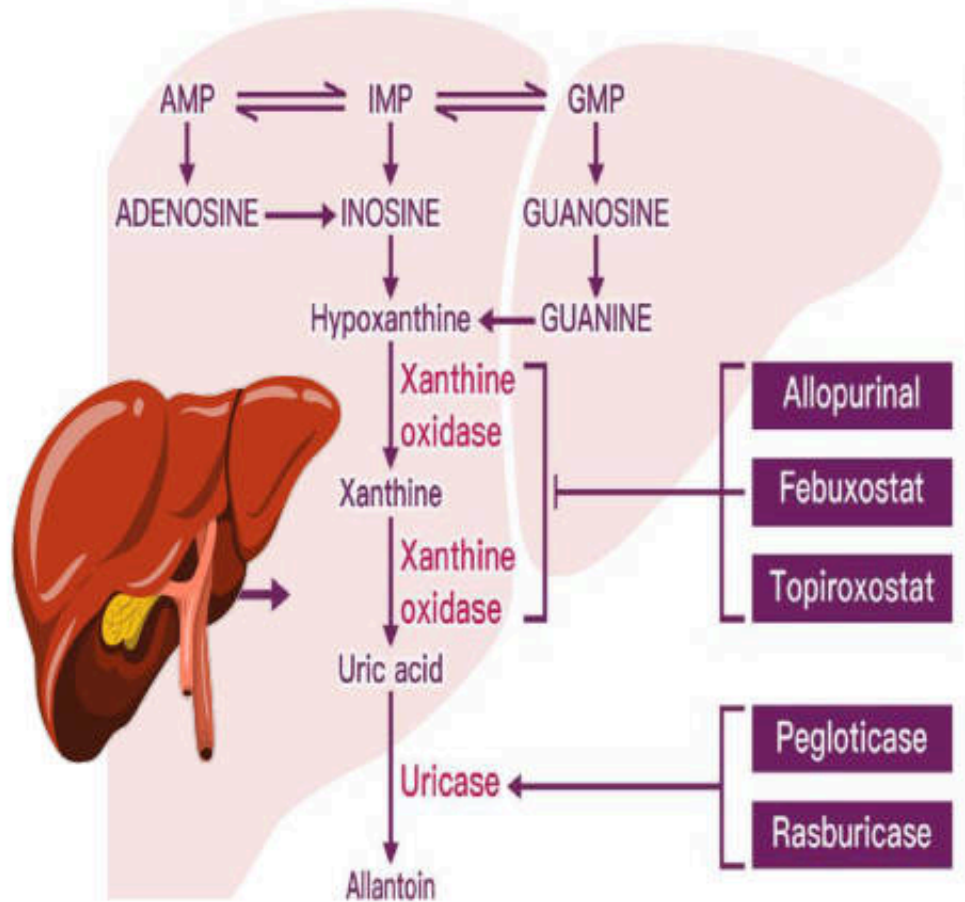
- Enzyme for degradation excess circulating uric acid
- Drugs: Rasburicase (withdrawn), **Pegloticase**
- For treatment of severe tophaceous gout or damaging arthropathy who have failed or are intolerant of conventional therapy

Urate-lowering therapeutic drugs approved in any country

Medication	MOA	Approval	Company	Benefits	Drawbacks
Allopurinol	X	FDA, EMA, PMDA	Generic	Inexpensive, well-studied	Severe cutaneous allergic reactions
Febuxostat	X	FDA, EMA, PMDA	Takeda	Superior SUA-lowering effect relative to allopurinol	Black-box warning for increased risk of cardiovascular death
Topiroxostat	X	PMDA	Sanwa Kagaku Kenkyusho, Fuji Yakuhin	Novel alternative to allopurinol	Limited availability, twice daily dosing
Benzbromarone	U	EMA, PMDA	Sanofi-Aventis	Comparable SUA effect as allopurinol, superior to probenecid	Hepatotoxicity, limited worldwide availability
Lesinurad	U	FDA	AstraZeneca	Selective activity on URAT1	Not approved for monotherapy, removed from US market
Probenecid	U	FDA, EMA, PDMA	Lannett	Well-tolerated, long history of use	Nephrolithiasis, drug-drug interactions
Sulfinpyrazone	U	FDA, EMA	Novartis	–	Acute kidney injury, removed from US market
Pegloticase	R	FDA	Horizon Therapeutics	No need to renally adjust dose	Infusion reactions, ADA, high cost

MOA, mechanism of action; X, xanthine oxidase inhibitor; U, uricosuric; R, recombinant uricase; FDA, US Food and Drug Administration; EMA, european medicine administration; SUA, serum uric acid; ADA, antidrug antibodies.

1. XANTHINE OXIDASE INHIBITORS (XOIS, URICOSTATIC)



First-line therapy

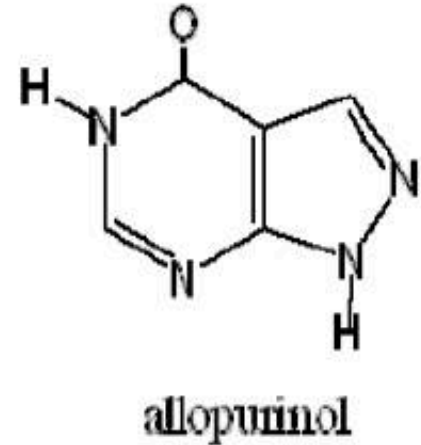
ALLOPURINOL



FEBUXOSTAT

**Alternative for
patients with
allopurinol allergies**

ALLOPURINOL



✓ **FIRST-LINE DRUG FOR TREATMENT**

✓ **Purine-based xanthine oxidases inhibitor**

✓ **"START SLOW GO SLOW"**

- starting daily doses of ≤ 100 mg and gradually titrate dose upto 900 mg

✓ **ADJUSTED DOSE IN PATIENTS WITH RENAL DISORDERS**

✓ **HYPERSENSITIVITY**

- 2–8% Allopurinol hypersensitivity syndrome (AHS): rash, eosinophilia, leukocytosis, fever, hepatitis and progressive kidney failure
- HLA-B*58:01 allele significantly increases the risk of Severe cutaneous adverse reactions (SCARs) in Han Chinese, Korean, Thai, and African descent
- Need to test for the **HLA-B*5801 allele** prior to starting allopurinol

Limitations



ONE-THIRD OF PATIENTS FAIL TO CONTROL SU levels (<6 mg/dl)



Strong association between *HLA-B*5801* and allopurinol-induced Stevens–Johnson syndrome and toxic epidermal necrolysis in a Thai population

Wichitra Tassaneeyakul;Thawinee Jantararoungtong;Pei Chen;Pao-Yu Lin;Somsak Tiamkao;Usanee Khunarkornsiri;Pachadaporn Chucherd;Parinya Konyoung;Suda Vannaprasaht;Charoen Choonhakarn;Pornrith Pisuttimarn;Alisara Sangviroon;Wongwiwat Tassaneeyakul;

Table 3 The association of individual *HLA-B* allele with allopurinol-induced SJS/TEN

<i>HLA-B</i> allele	Allopurinol-induced SJS/TEN (n=27)	Allopurinol tolerant (n=54)	OR	95% CI	P value
5801	27 (100.0)	7 (13.0)	348.3	19.2–6336.9	1.61×10^{-13}
1301	4 (14.8)	10 (18.5)	0.8	0.2–2.7	0.68
1502	2 (7.4)	10 (18.5)	0.4	0.1–1.7	0.18
1802	3 (11.1)	2 (3.7)	3.3	0.5–20.7	0.19
3802	1 (3.7)	5 (9.3)	0.4	0.04–3.4	0.37
4001	1 (3.7)	7 (13.0)	0.3	0.03–2.2	0.19
4601	3 (11.1)	16 (29.6)	0.3	0.08–1.1	0.06
5101	1 (3.7)	8 (14.8)	0.2	0.03–1.9	0.13

เปิดตรวจยืนยันแพ้ยาโรคเกาต์

ไทยรัฐฉบับพิมพ์

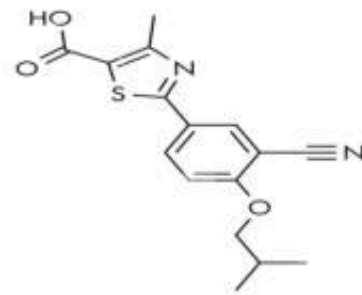
31 ต.ค. 2562 09:15 น.

1.ชุดตรวจยืนยัน HLA-B*58:01 ก่อนให้ยาลดกรดยูริก อัลโลพูรินอล หรือยารักษาโรคเกาต์ 2.ชุดตรวจยืนยัน HLA-B*57:01 ก่อนให้ยาด้านไวรัสเชไอวี อะบาคาเวียร์ และ 3.ชุดตรวจยืนยัน HLA-B*15:02 ก่อนให้ยากันชักคาร์บามาซีปีน อย่างไรก็ตาม การตรวจยืนยันแพ้ยามีความสำคัญ เพราะพบว่า อัตราการพบยืนยันแพ้ยาในคนไทยอยู่ที่ร้อยละ 15 หรือ 9.7 ล้านคน พบในผู้ป่วย 42,000-100,000 คน ซึ่งการตรวจยืนยันก่อนให้ยาจะลดการเกิดผื่นแพ้ยารุนแรง การแพ้ยา และลดอัตราตายได้มากกว่าร้อยละ 90 ช่วยประหยัดงบประมาณจากอาการแพ้ยารุนแรงปีละกว่า 250 ล้านบาท

นพ.โอภาสกล่าวว่า ทั้งนี้เพื่อมอบเป็นของขวัญปีใหม่ปี 2563 ให้กับคนไทย โดยกรมจะมีการเปิดให้บริการตรวจยืนยันแพ้ยารักษาโรคเกาต์หรือยาลดกรดยูริก ในวันที่ 1 ม.ค. 2563 เป็นต้นไป ซึ่งจะดำเนินการตรวจฟรี 10,000 คน โดยรับการตรวจได้ที่กรมวิทยาศาสตร์การแพทย์และศูนย์วิทยาศาสตร์การแพทย์อีก 14 แห่งทั่วประเทศคือ เชียงใหม่ เชียงราย พิชณุโลก นครสวรรค์ สมุทรสงคราม ชลบุรี ขอนแก่น อุตรธานี นครราชสีมา อุบลราชธานี สุราษฎร์ธานี ตรัง สงขลา และภูเก็ต ซึ่งการตรวจยืนยันแพ้ยารักษาโรคเกาต์ไม่ต้องเสียค่าใช้จ่าย เพราะอยู่ในสิทธิประโยชน์ทั้ง 3 กองทุนสุขภาพภาครัฐ คือ ระบบหลักประกันสุขภาพถ้วนหน้า หรือบัตรทอง สวัสดิการข้าราชการ และสิทธิประกันสังคม ทั้งนี้ ในส่วนของการตรวจยืนยันแพ้ยารักษาโรคเกาต์ จะช่วยประหยัดค่าใช้จ่ายทางการแพทย์ของประเทศได้มากกว่า 90 ล้านบาทต่อปี.

Febuxostat

- Non-purine xanthine oxidase inhibitor
- Initial dose: 20–80 mg PO qDay; increase to 120 mg PO q Day after 2 wk. if serum uric acid <6 mg/dL is not achieved
- 65% of patients achieved serum uric acid levels less than 6.0 mg/dL
- Predominantly clearance via hepatic metabolism
- **Side effects:** muscle pain, gastrointestinal discomfort, skin rashes, diarrhea, and a slight elevation in liver enzymes



febuxostat

Key message

More potent reduces serum uric acid than allopurinol but more expensive

Alternative for patients with allopurinol allergies

concerns regarding the cardiac safety profile

1. Xanthine oxidase inhibitors (XOIs)

Characteristic	Allopurinol	Febuxostat
Mechanism of actions	purine analogue, nonspecific competitive inhibitor of xanthine oxidase Active metabolite "oxypurinol"	non-purine, <u>highly specific XO</u>
Elimination	Renal	Hepatic
Daily dose*	(50 mg), 100–900 mg (300 mg daily): " <u>start slow, go slow</u> "	40-80 mg, 120 mg
Hypersensitivity syndrome: DRESS and SCAR	"Rare" <u>HLA-B*5801</u> allele: Korean, Chinese and Thai populations	"Rare" "initially thought to be free of the serious SCAR and DRESS but severe hypersensitivity reactions have subsequently been reported"
Cautions	Severe renal impairment	Heart failure (NYHA class III or IV), hepatic failure, eGFR<30ml/min/1.73m ²
Drug Interactions	Cytotoxic drugs metabolized by XO (for example, azathioprine, mercaptopurine), Warfarin	Cytotoxic drugs metabolized by XO (for example, azathioprine, mercaptopurine)

Limitations of the current uricostatic drugs

current antihyperuricemic drugs

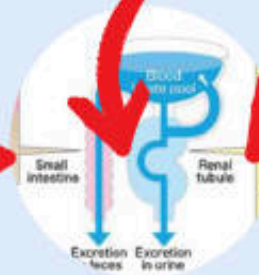
Allopurinol

- Several adverse reactions (ADRs): GI effects, rash, and Stevens–Johnson’s syndrome
- Contraindications in renal impairment

Fail to reach the target serum urate levels (6 mg/dl)

Febuxostat

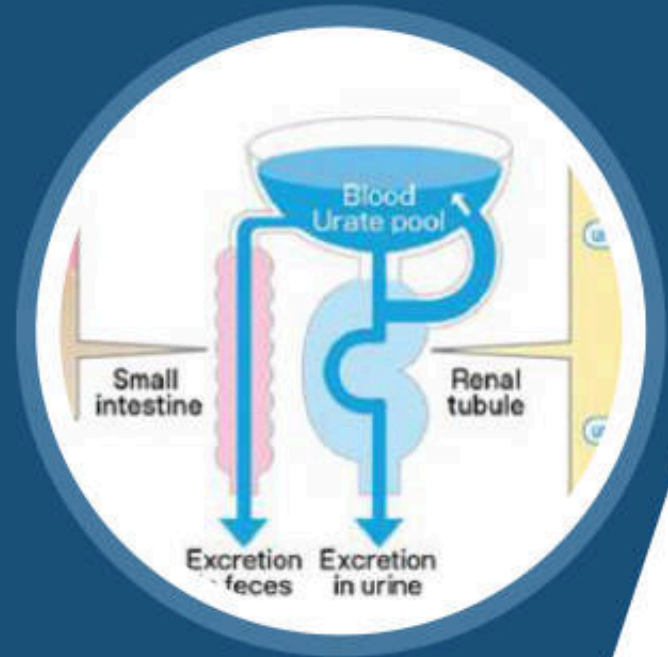
Expensive and the concern of cardiovascular events



Safer new drugs are needed

2. URICOSURIC DRUGS

(ยาเร่งการขับกรดยูริก
ทางไต)



...

Uricosuric agents

Mechanism of actions

- Increase renal uric acid excretion by inhibiting urate-reabsorbing transporters, particularly URAT1, in the proximal tubular cells of the kidneys
- Usually reserved for patients who cannot tolerate XOIs, or for combination use in patients with inadequate response to XOIs monotherapy

ADRs

- Higher concentration of uric acid in the collecting tubules can predispose to uric acid stone formation
- Patient should be advised to drink plenty of fluids

Drugs

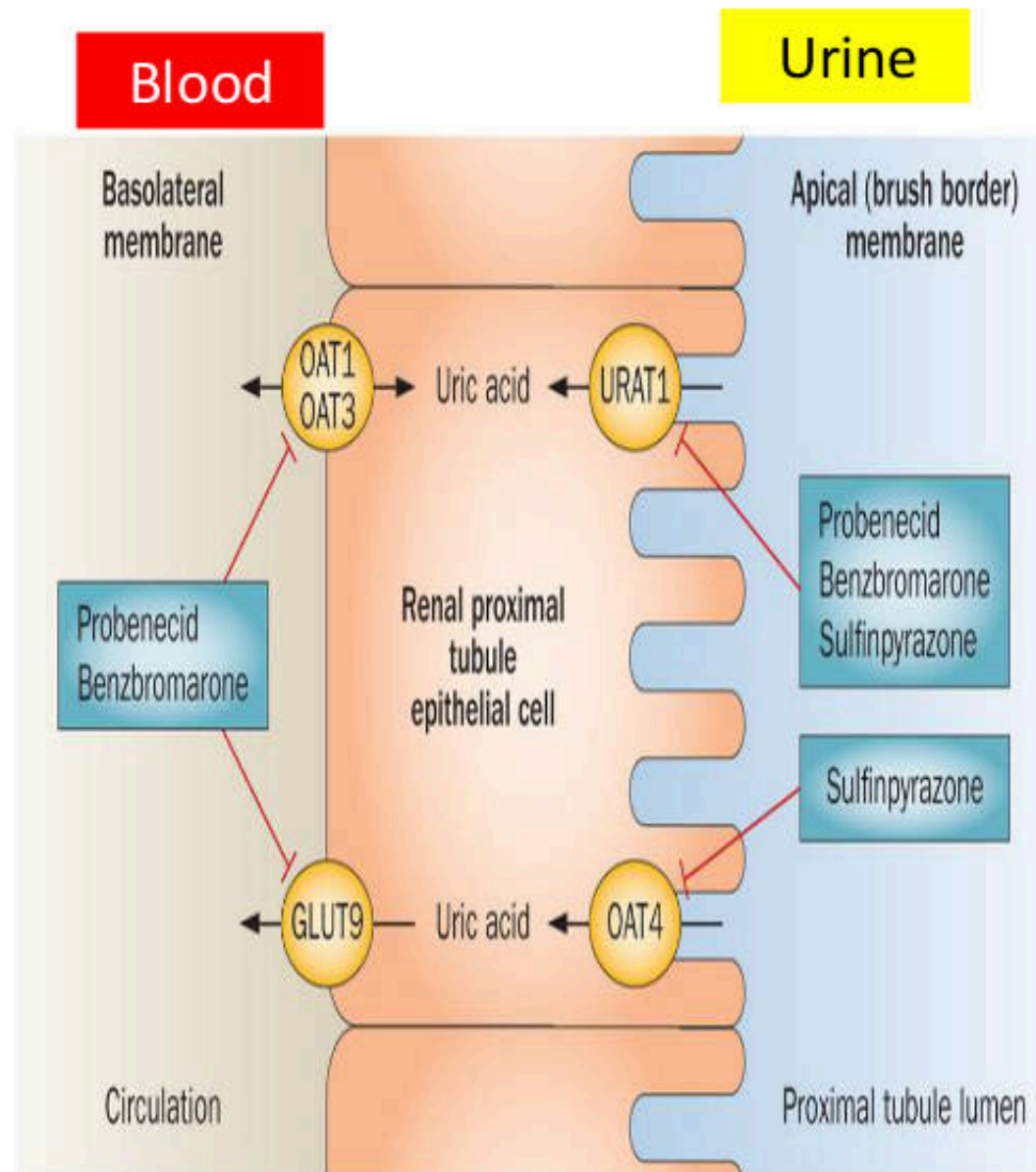
- Benzbromarone (50–200 mg daily)
- Probenecid (250– 500 mg twice daily)
- Sulfapyrazone (200–800 mg daily)
- Lesinurad: new drug approved in 2015 but was withdrawn
- Dotinurad: new drug approved in 2023

“Probenecid”

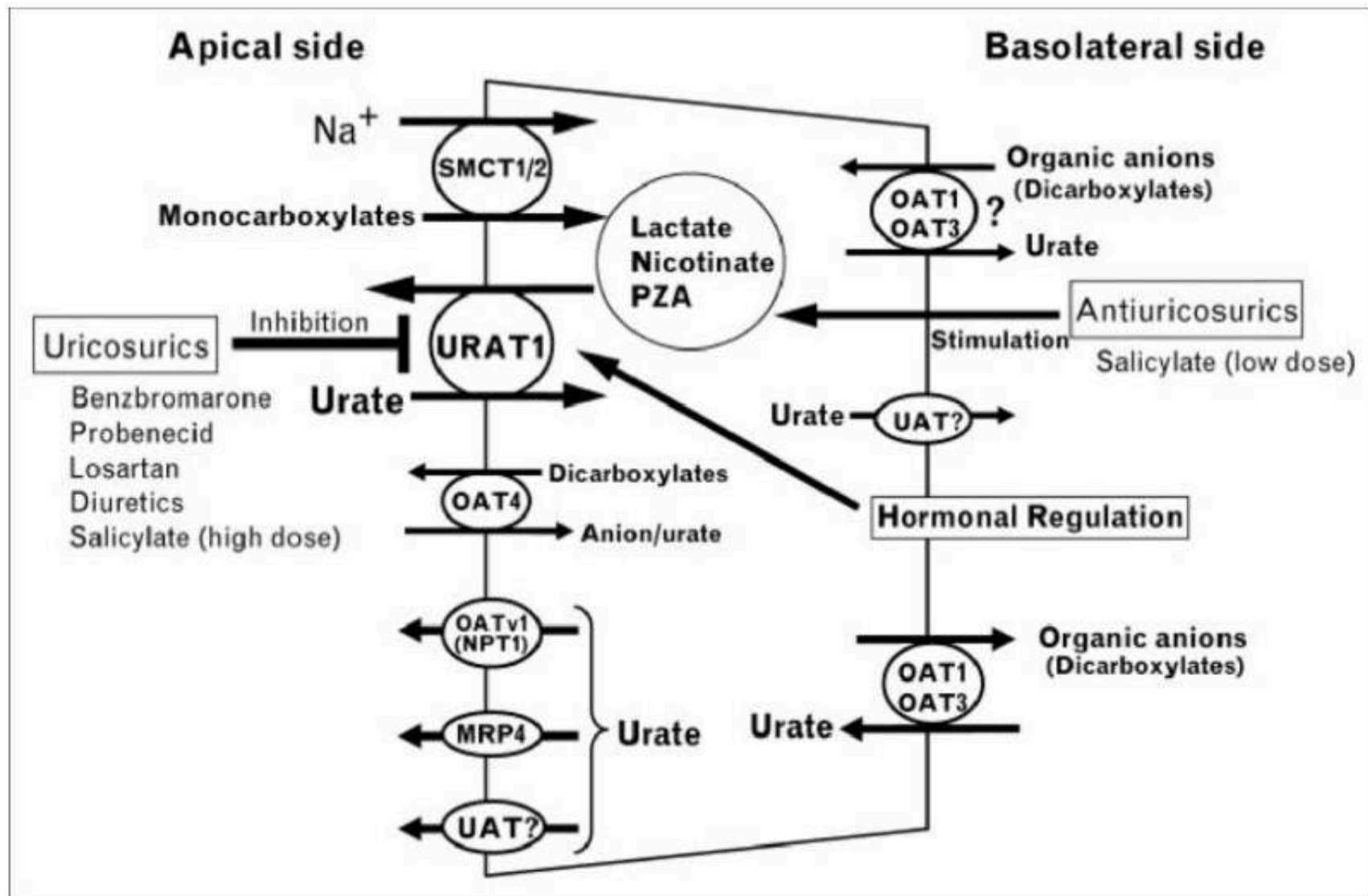
- Inhibit URAT1, OAT1 and OAT3 (drug interactions)
- AEs: impaired with decreasing glomerular filtration rate (GFR)
- Efficacy and safety in gout patients is extremely limited

“Benzbromarone”

- Inhibit URAT 1 and GLUT 9 transporters
- more potent than probenecid
- not contraindicated in patients with mild or moderate renal impairment
- postmarketing reports of **abnormal liver function** and deaths from hepatic failure
→→→→ withdrawn in some country!!



Available uricosuric drugs acting at proximal renal tubule

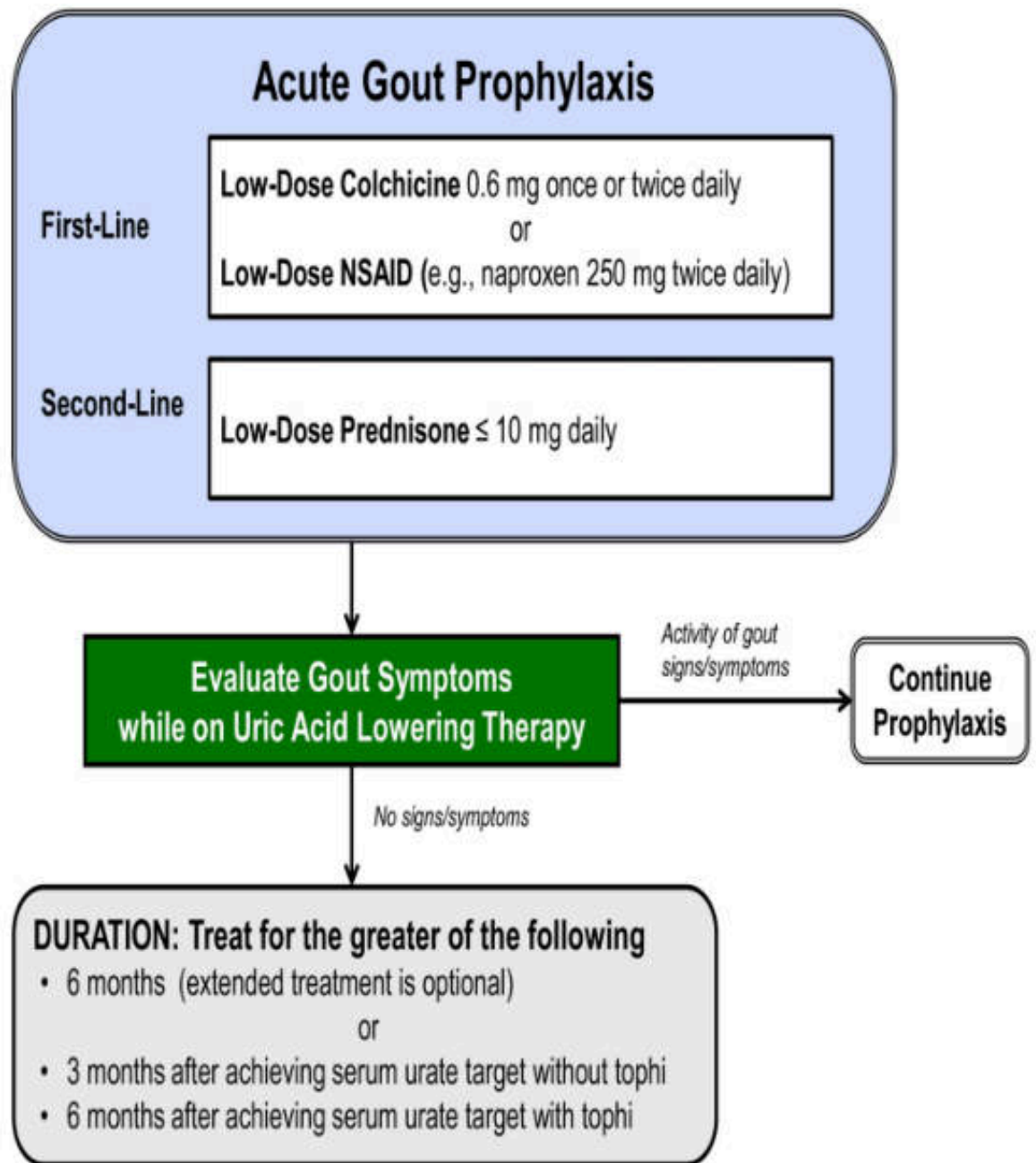


Precaution of uricosurics

- Checking urinary uric acid level for patients receiving uricosuric treatment
- Patients with known renal calculi or moderate-to-severe CKD (stage >3) should not be treated with uricosurics
- Receive counseling about adequate hydration (>1 L)
- **Alkalinizing the urine is conditionally recommended *against* for patients receiving uricosuric treatment**

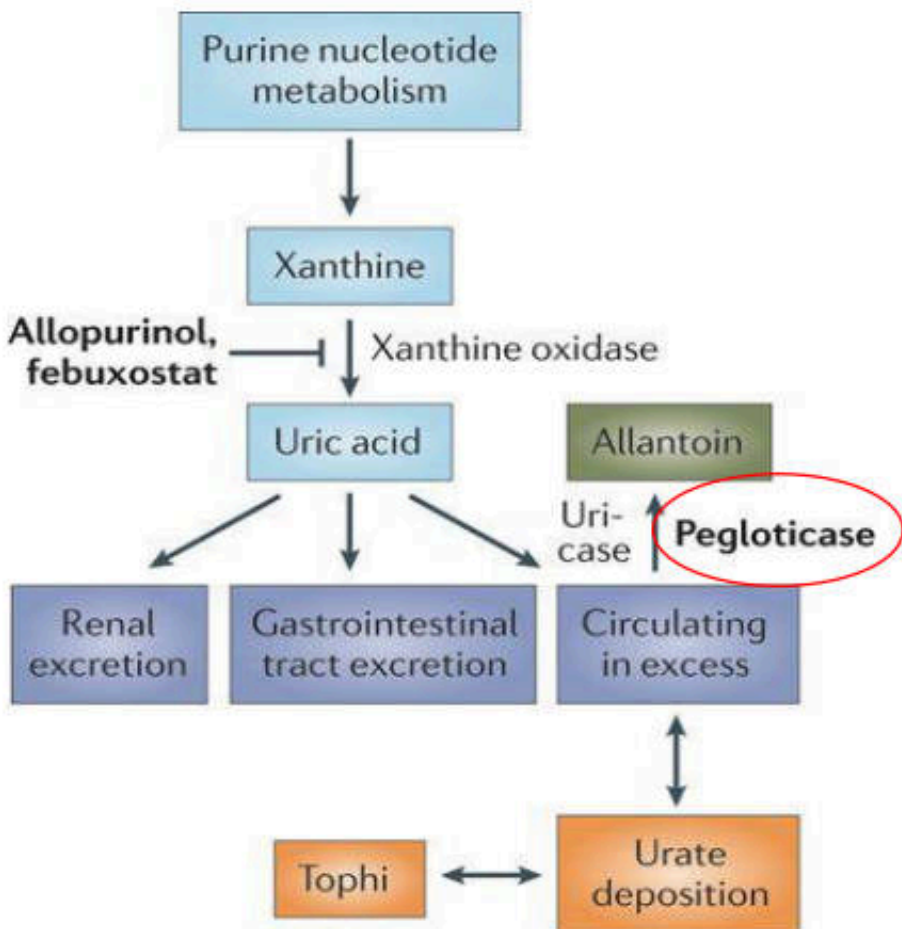
Other available ULTs: Losartan, oral vitamin C supplements, and fenofibrate

?เหตุใดจำเป็นต้องใช้
ยาป้องกันอาการปวด
เช่น colchicine
หรือ NSAIDs
ร่วมกับใช้ยา
allopurinol
ในช่วง 3-6 เดือนแรก



3. Uricase: Pegloticase

For treatment of refractory gout and **use over continuing current ULT for patients with gout for whom XO1 treatment, uricosurics, and other interventions have failed to achieve the SU target**



- Recombinant uricase expressed in E.coli
 - Enzyme that lowers the levels of uric acid by catalyzing the oxidation of uric acid to allantoin
 - Uricase speeds resolution of tophi

- Adverse events
 - Anaphylaxis, 6.5%
 - Infusion reactions, 26%
 - Gout flares,

- Patients should be pre-medicated
 - Antihistamines and corticosteroids
 - Gout prophylaxis

Reduction in SUA <6mg/dl			
Treatment	GOUT 1	GOUT 2	Combined
8mg q 2 weeks	46.5%	38.1%	42.3%
8 mg q 4 weeks	19.5%	48.8%	34.5%
Placebo	0	0	0

Treatment	Flares	Tophus
8mg q 2 weeks	77%	40.4%
8 mg q 4 weeks	81%	21.2%
Placebo	54%	0

ADRs of pegloticase

Table 1. Documented adverse events (AEs) with pegloticase in a phase III trial and open-label extension.

Adverse events	Range of reported AEs in a phase III trial and OLE
Gout flare	71–85%*
Infusion reactions: monthly pegloticase	42%
Infusion reactions: biweekly pegloticase	26%
Headache	9–11%*
Nausea	7–12%*
Back pain	4–17%*
Nasopharyngitis	5–10%*
Cardiovascular events: congestive heart failure, arrhythmia, unstable angina, deep venous thrombosis, transient ischemic attack, coronary revascularization	2–7%*
Cardiovascular death, nonfatal MI	1–2%*

Data obtained from trials by Sundy *et al.*¹⁹ and Becker *et al.*³⁵

- Pegloticase has the potential for inciting immunogenicity.
- Some patients develop antipegloticase antibodies with accompanying loss of efficacy and potential development of infusion reactions (IRs)

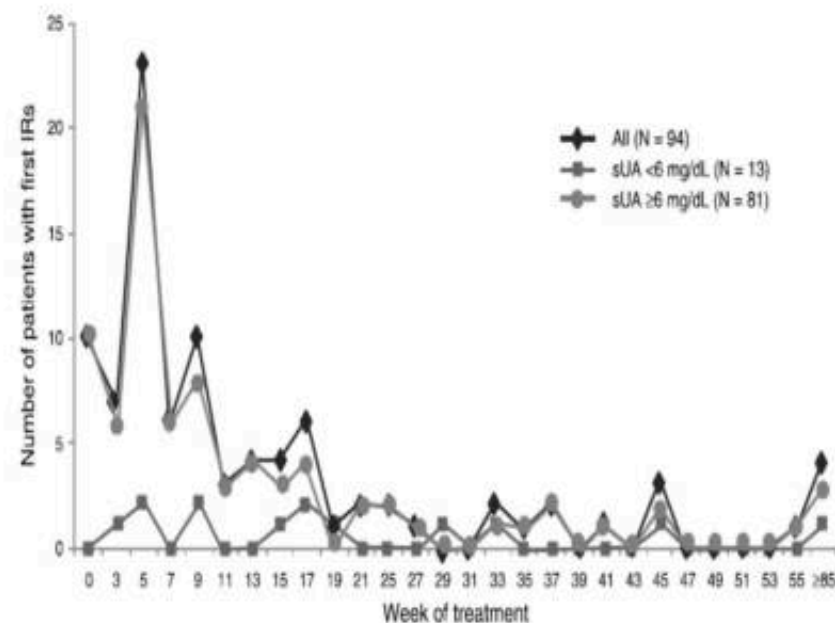


FIGURE 1. Number of first IRs with pegloticase by treatment week and sUA.

Summary

- ❖ Lowering serum uric acid to a target level(<6 mg/dl)
- ❖ **Allopurinol**: 1st-line drug but target SUA concentrations are not always achieved and may cause severe adverse event: Allopurinol Hypersensitivity Syndrome (AHS)
- ❖ **Febuxostat**: high efficacy but high cost
- ❖ **Combination of uricosurics and XOIs**: reduce sUA in chronic gout but safety in patients with CKD and other comorbidities is still unknown
- ❖ **Pegloticase**: gout with refractory to conventional ULT or patients with severe disease burden but is limited by injection reactions, high cost, or formation of neutralizing antipegloticase antibodies

แบบฝึกหัดเสริมความเข้าใจ

1. ยาชนิดใดอาจก่อให้เกิดการแพ้อย่างรุนแรงในผู้ป่วยที่มียีน

HLAB*5801

- A. Febuxostat
- B. Pegloticase
- C. Allopurinol
- D. Canakinumab

2. ข้อใดเป็นกลไกการออกฤทธิ์ยา **pegloticase**

- A. xanthine oxidase inhibition
- B. URAT1 inhibition
- C. IL-1 inhibition
- D. mimic Uricase

อ่านรายละเอียดเพิ่มเติมในเอกสารคำสอนเรื่องเภสัชวิทยาและหลักการ รักษาภาวะกรดยูริกในเลือดสูงและโรคเกาต์



Thank you